

Review

Summary

This paper presents **TrustFedGNN**, a federated graph learning framework that jointly addresses fairness, privacy, robustness, and explainability for high-risk applications. The framework combines fairness-aware graph learning, differential privacy, robust aggregation, and trust assessment, and is evaluated on multiple datasets against a large number of baselines.

Overall, the paper is well written, the experiments are comprehensive, and the conclusions are generally supported by the reported results. The authors also discuss the limitations of their work, which I appreciate. I recommend **minor revisions** before publication.

Comments

1. **Section 5.8.** The statement that *robust-bfwa is the only aggregator that stays strong under all three attacks* is stronger than what Table 9 shows. The trimmed-mean aggregator achieves comparable utility and even performs better under the ALIE attack. I suggest softening this claim.
2. **Number of baselines.** The manuscript refers to **sixteen baselines**, whereas the main comparison tables appear to contain fifteen. Please check whether one baseline is missing from the tables or revise the reported number throughout the manuscript.
3. **Baseline comparisons across datasets.** The paper gives the impression that all datasets are evaluated against the full set of baselines. However, the Credit and especially the Elliptic datasets include fewer competing methods. This should be stated explicitly in the experimental setup.
4. **Robustness experiments.** Since robustness is one of the main contributions, the conclusions are currently based on single-seed experiments. Adding a few additional runs would strengthen the evidence. If this is not feasible, the claims should be slightly moderated.
5. **Energy consumption results.** The energy column in Table 11 reports 0.00 Wh for all methods, making it difficult to draw any comparison. Either report the values with higher precision or clarify that the measurements fall below the resolution of the measurement tool.
6. I also noticed a few inconsistencies between the text and tables that should be corrected:
 - The reported FedFACT DPD value on the Bail dataset differs between the text and Table 5.
 - The reported p-value for FairFed on the German dataset is inconsistent.
 - The statement that all non-private baselines significantly outperform the private model in AUC does not appear to hold for FairGNN.
 - The attribute inference baseline is reported as both 0.50 and 0.53 in different parts of the manuscript.
 - There is a small discrepancy between the German DPD value reported in the ablation study and the corresponding main results table.
7. Finally, the abstract and the description of the differential privacy guarantee should make it clearer that the guarantee applies to the **released fairness statistics**, rather than the entire model update.

Recommendation

Minor Revision

The paper presents a useful and timely contribution, and the experimental evaluation is generally convincing. The issues raised above are mostly related to consistency, wording, and clarification rather than the methodology itself, and I believe they can be addressed without additional major experiments.